

Derivation of the Disturbance Equations

From the compressible Navier–Stokes equations to the
disturbance eigenvalue problem solved by `pyMack`

`pyMack` technical documentation

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Contents

1	Scope	1
2	Linearisation and the normal-mode ansatz	3
3	The four disturbance equations	4
4	Operator form and the discretised derivative	5
5	Physical character of the instability modes	7
6	Assumptions and range of validity	9

1 Scope

Linear stability theory (LST) determines whether an infinitesimal wavelike disturbance superposed on a steady laminar boundary layer amplifies or decays as it propagates [1]. Because the disturbance is infinitesimal, products of disturbance quantities are negligible and its governing equations are linear; because the coefficients of the linearised equations depend, under the parallel-flow assumption, only on the wall-normal coordinate y , the normal-mode substitution $q' = \hat{q}(y) e^{i(\alpha x - \omega t)}$ removes x and t from the problem entirely. The requirement that a non-trivial shape $\hat{q}(y)$ exist then constitutes an eigenvalue problem.

This document derives that eigenvalue problem. Section 2 linearises the governing equations and introduces the normal-mode ansatz; Section 3 states the four resulting ordinary differential equations exactly as assembled in `pymack/temporal_solver.py`; Section 4 collects them into a compact operator form, the intermediate step between the present derivation and the numerical implementation, and defines the transition from the continuous derivative to the discretised derivative operator; Section 5 summarises the physical character of the instability modes the equations support; Section 6 records the idealisations underlying the derivation. The mean-flow profiles that enter as coefficients are computed in the companion document *The Mean Boundary-Layer Profile* and are treated here as known data; the discretisation and eigenvalue algorithms are the subject of the companion *Numerical Methods* document. Figure 1 places these stages in the complete computational sequence.

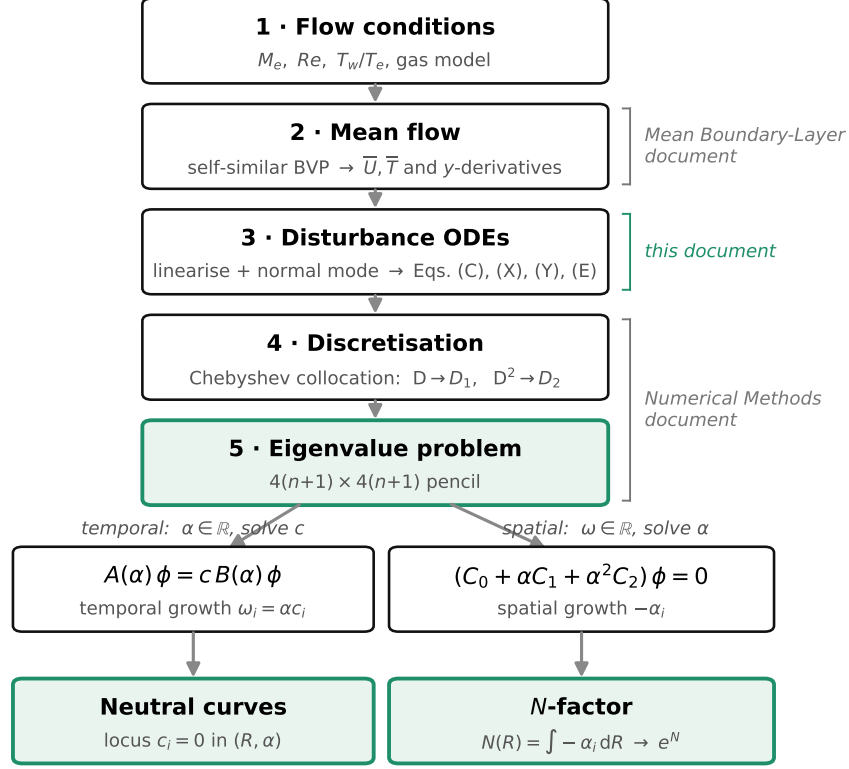


Figure 1: The complete LST computation in pyMack. Stages 1–2 are covered by the companion *Mean Boundary-Layer* document, stage 3 by the present document, and stages 4–5 by the companion *Numerical Methods* document; the two solution routes are temporal (fixed real α , complex c) and spatial (fixed real ω , complex α).

Notation. Overbars denote mean (base-flow) quantities; hats denote complex disturbance amplitudes. The continuous wall-normal amplitude vector is $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$. A prime on a mean quantity denotes $D \equiv d/dy$; from Section 4 onward the distinction between the continuous operator D and its discrete counterpart is made explicit.

Base flow & transport		Disturbance / eigenvalue	
$\bar{U}, \bar{T}, \bar{\rho}=1/\bar{T}$	mean velocity, temp., density	$\hat{q}$	amplitude shape, Eq. (6)
$\bar{\mu}, \bar{\kappa}$	viscosity, conductivity	$\hat{u}, \hat{v}, \hat{T}, \hat{p}$	velocity/temp./pressure amplitudes
M_e, Re, Pr, γ	Mach, Reynolds, Prandtl, c_p/c_v	α, ω	wavenumber, frequency
$\nu_R=1/(Re\bar{\rho})$	viscous prefactor	$c=\omega/\alpha$	complex phase speed; c_r, c_i its parts
$\mathcal{C}_\kappa, \mathcal{C}_d, \mathcal{L}_c$	conduction/dissipation groups	M_{rel}	relative Mach number, §5

2 Linearisation and the normal-mode ansatz

The flow is governed by the compressible Navier–Stokes equations, expressing conservation of mass, momentum, and energy for a viscous, heat-conducting perfect gas [2]:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0, \quad (\text{mass}) \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau}, \quad (\text{momentum}) \quad (2)$$

$$\rho c_p \frac{DT}{Dt} = \frac{Dp}{Dt} + \nabla \cdot (\kappa \nabla T) + \Phi, \quad (\text{energy}) \quad (3)$$

$$p = \rho RT, \quad (\text{state}) \quad (4)$$

where $\mathbf{u} = (u, v)$ is the velocity, ρ the density, T the temperature, p the pressure, μ the viscosity, κ the conductivity, c_p the specific heat, R the gas constant, $D/Dt = \partial_t + \mathbf{u} \cdot \nabla$ the material derivative, $\boldsymbol{\tau}$ the Newtonian viscous stress tensor, and $\Phi = \boldsymbol{\tau} : \nabla \mathbf{u}$ the dissipation function (both linear in the velocity gradients). These are the same equations solved for the steady mean flow in the companion document; they are applied here a second time, in linearised form, for the disturbance. The reduction to ordinary differential equations proceeds in five steps.

Step 1: decomposition into mean and disturbance. Each field is split as

$$u = \bar{U}(y) + u', \quad v = v', \quad T = \bar{T}(y) + T', \quad p = \bar{p} + p', \quad \rho = \bar{\rho}(y) + \rho', \quad |\cdot'| \ll |\cdot|, \quad (5)$$

with $\bar{v} = 0$ for a parallel mean flow. The collection $q' = (u', v', T', p', \rho')$ is the disturbance.

Step 2: linearisation. Substituting (5) into (1)–(4), the mean satisfies the equations by itself and cancels; terms quadratic in the primed quantities are dropped. The remaining equations are linear in q' with coefficients constructed from the known mean profiles.

Step 3: parallel-flow assumption. The boundary layer grows slowly, so locally $\bar{q} = \bar{q}(y)$ only and the coefficients depend on y alone. This neglects the $O(1/R)$ streamwise growth of the mean flow and of the eigenfunction; the systematic remedy, retaining that slow x -variation, is the parabolised stability equations [8] (cf. Section 6).

Step 4: normal-mode ansatz. The disturbance is represented as a single travelling wave,

$$q'(x, y, t) = \hat{q}(y) e^{i(\alpha x - \omega t)}, \quad \hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})^\top, \quad (6)$$

with α the streamwise wavenumber, ω the frequency, and $c = \omega/\alpha$ the complex phase speed. Substitution replaces $\partial_x \mapsto i\alpha$, $\partial_t \mapsto -i\omega$, and $\partial_y \mapsto D$, collapsing the linear PDEs to coupled ODEs in y (Figure 2). The spanwise wavenumber is set to zero ($\beta = 0$); a general oblique wave $\propto e^{i(\alpha x + \beta z - \omega t)}$ propagates at angle $\psi = \arctan(\beta/\alpha)$, and since Squire’s theorem does not carry over to compressible flow, the first mode is most unstable oblique ($\psi \sim 55^\circ$) while the second mode is most unstable at $\psi = 0$ [1]. The consequences of the two-dimensional restriction are recorded in Section 6.

Step 5: elimination of density. The linearised equation of state gives

$$\frac{\rho'}{\bar{\rho}} = \gamma M_e^2 \hat{p} - \frac{\hat{T}}{\bar{T}}, \quad (7)$$

which removes ρ' from the system. Four amplitude fields remain, $\hat{u}, \hat{v}, \hat{T}, \hat{p}$, with $\hat{p}$ scaled by $\rho_e U_e^2$ (so that $\gamma M_e^2 \hat{p} = p'/p_e$).

A worked linearisation. To make the mechanics explicit, consider mass conservation (1), $\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$. Inserting (5), the steady mean satisfies $\partial_x(\bar{\rho} \bar{U}) = 0$ and cancels; retaining only terms linear in the primed quantities,

$$\partial_t \rho' + \bar{U} \partial_x \rho' + \bar{\rho} \partial_x u' + \partial_y(\bar{\rho} v') = 0. \quad (8)$$

Applying the ansatz (6) ($\partial_t \rightarrow -i\alpha c$, $\partial_x \rightarrow i\alpha$, $\partial_y \rightarrow D$), substituting ρ' from (7), collecting terms, and dividing by $\bar{\rho}$ yields exactly Eq. (9) below. The remaining three equations follow by the identical procedure applied to (2)–(3); the algebra is mechanical but lengthy, and only the result is printed.

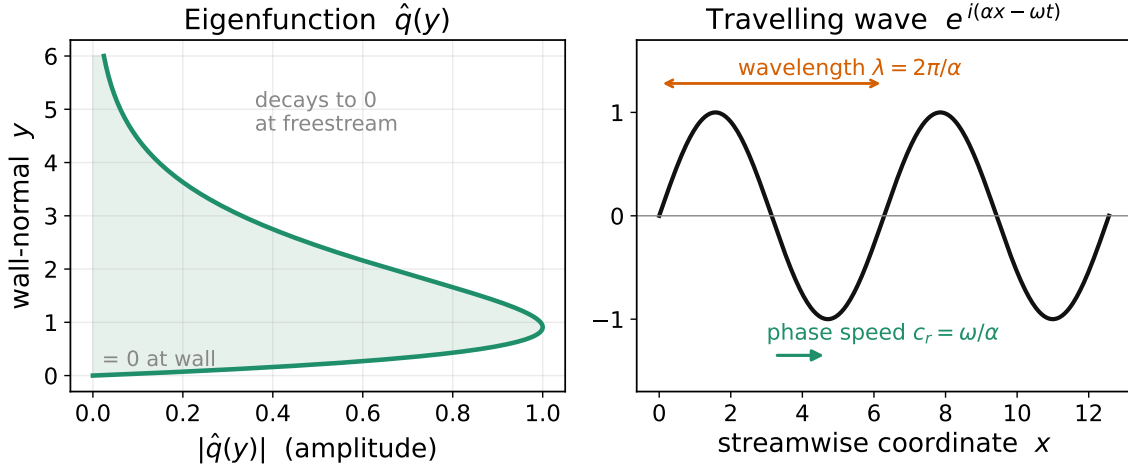


Figure 2: The ansatz (6) factorises a disturbance $q' = \hat{q}(y) e^{i(\alpha x - \omega t)}$ into a wall-normal *shape* $\hat{q}(y)$ (left; the eigenfunction, zero at the wall and decaying at the edge) and a travelling wave in x (right; wavelength $2\pi/\alpha$, phase speed $c_r = \text{Re}(c)$).

3 The four disturbance equations

The linearised system consists of four coupled ODEs in y for $\hat{u}, \hat{v}, \hat{T}, \hat{p}$: continuity (with the equation of state folded in through (7)), streamwise and wall-normal momentum, and energy. Fixing one of α or ω real leaves the other (through c or α) as the eigenvalue. Every overbarred quantity is a known base-flow coefficient; all transport derivatives ($d\mu/dT$, $d\kappa/dT$, ...) are evaluated at the local mean temperature $\bar{T}(y)$ and are therefore known functions of y . The form below, assembled in `pymack/temporal_solver.py`, follows Özgen & Kırcah [3]; it is equivalent to the canonical compressible stability equations of Mack [1] and Malik [4] restricted to two dimensions. Momentum and energy are divided by $\bar{\rho}$, and the viscous prefactor is written $\nu_R \equiv 1/(Re \bar{\rho})$; every $\bar{\rho}$ is the base-flow density $\bar{\rho}(y) = 1/\bar{T}(y)$, uniformly inside ν_R , $\mathcal{C}_\kappa$, and $\mathcal{C}_d$.

(C) Continuity + equation of state.

$$\mathrm{i}\alpha\hat{u} + \mathrm{D}\hat{v} - \frac{\bar{T}'}{\bar{T}}\hat{v} + \mathrm{i}\alpha(\bar{U} - c)\left(\gamma M_e^2\hat{p} - \frac{\hat{T}}{\bar{T}}\right) = 0. \quad (9)$$

(X) Streamwise momentum.

$$\begin{aligned} \mathrm{i}\alpha(\bar{U} - c)\hat{u} + \bar{U}'\hat{v} + \mathrm{i}\alpha\bar{\rho}^{-1}\hat{p} = \nu_R \Big[\bar{\mu}(\mathrm{D}^2 - \frac{4}{3}\alpha^2)\hat{u} + \bar{\mu}'\mathrm{D}\hat{u} + \mathrm{i}\alpha(\frac{1}{3}\bar{\mu}\mathrm{D} + \bar{\mu}')\hat{v} \\ + \frac{\mathrm{d}\bar{\mu}}{\mathrm{d}T}(\bar{U}'' + \bar{U}'\mathrm{D})\hat{T} + \frac{\mathrm{d}^2\bar{\mu}}{\mathrm{d}T^2}\bar{U}'\bar{T}'\hat{T} \Big]. \end{aligned} \quad (10)$$

(Y) Wall-normal momentum.

$$\mathrm{i}\alpha(\bar{U} - c)\hat{v} + \bar{\rho}^{-1}\mathrm{D}\hat{p} = \nu_R \Big[\bar{\mu}(\frac{4}{3}\mathrm{D}^2 - \alpha^2)\hat{v} + \frac{4}{3}\bar{\mu}'\mathrm{D}\hat{v} + \mathrm{i}\alpha(\frac{1}{3}\bar{\mu}\mathrm{D} - \frac{2}{3}\bar{\mu}')\hat{u} + \mathrm{i}\alpha\frac{\mathrm{d}\bar{\mu}}{\mathrm{d}T}\bar{U}'\hat{T} \Big]. \quad (11)$$

(E) Energy (temperature).

$$\mathrm{i}\alpha(\bar{U} - c)\hat{T} + \bar{T}'\hat{v} + (\gamma - 1)\bar{\rho}^{-1}(\mathrm{i}\alpha\hat{u} + \mathrm{D}\hat{v}) = \mathcal{C}_\kappa\mathcal{L}_c\hat{T} + \mathcal{C}_d \Big[2\bar{\mu}\bar{U}'(\mathrm{D}\hat{u} + \mathrm{i}\alpha\hat{v}) + \frac{\mathrm{d}\bar{\mu}}{\mathrm{d}T}(\bar{U}')^2\hat{T} \Big], \quad (12)$$

$$\mathcal{L}_c = \mathrm{D}^2 - \alpha^2 + \frac{1}{\bar{\kappa}}\frac{\mathrm{d}\bar{\kappa}}{\mathrm{d}T}(2\bar{T}'\mathrm{D} + \bar{T}'') + \frac{1}{\bar{\kappa}}\frac{\mathrm{d}^2\bar{\kappa}}{\mathrm{d}T^2}(\bar{T}')^2, \quad \mathcal{C}_\kappa = \frac{\gamma\bar{\mu}}{Pr Re \bar{\rho}}, \quad \mathcal{C}_d = \frac{\gamma(\gamma - 1)M_e^2}{Re \bar{\rho}}. \quad (13)$$

Structure of the equations. The following observations locate the physics in (9)–(12):

- $\mathrm{i}\alpha(\bar{U} - c)$ is convection by the mean flow relative to the wave; it vanishes at the *critical layer* $\bar{U} = c_r$, where the inviscid form of the equations becomes singular, which is why that layer governs much of the instability physics [1].
- D, D^2 are wall-normal derivatives; the $-\alpha^2$ terms arise from $\partial_x^2 \mapsto -\alpha^2$.
- The $\nu_R = 1/(Re\bar{\rho})$ terms are viscous; the factors $\frac{4}{3}, \frac{1}{3}$ correspond to the Stokes hypothesis for the bulk viscosity (the companion *Numerical Methods* document generalises this to a ratio $\lambda/\mu = 1.2$ in the spatial closure).
- $(\gamma - 1)\bar{\rho}^{-1}(\mathrm{i}\alpha\hat{u} + \mathrm{D}\hat{v})$ in (12) is pressure (dilatation) work: $\mathrm{i}\alpha\hat{u} + \mathrm{D}\hat{v}$ is the linearised velocity divergence, so compression or expansion of a fluid parcel heats or cools it, coupling the velocity field back into the temperature equation.
- The $\mathrm{d}\bar{\mu}/\mathrm{d}T, \mathrm{d}\bar{\kappa}/\mathrm{d}T$ terms represent temperature-dependent transport, essential at hypersonic conditions.
- $\mathcal{C}_\kappa\mathcal{L}_c\hat{T}$ is heat conduction; $\mathcal{C}_d[\dots]$ is viscous dissipation.

4 Operator form and the discretised derivative

Equations (9)–(12) are linear both in the amplitude vector $\hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})^\top$ and in the eigenvalue, and every derivative they contain is of at most second order. They can therefore be collected into the single operator statement

$$\mathbf{L}(\alpha; y, \mathrm{D}) \hat{q} = c \mathbf{M}(\alpha; y) \hat{q}, \quad \mathbf{L} = \mathbf{A}_2(y) \mathrm{D}^2 + \mathbf{A}_1(\alpha; y) \mathrm{D} + \mathbf{A}_0(\alpha; y), \quad (14)$$

with rows ordered (C, X, Y, E) and columns ordered $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$. Two structural facts make (14) particularly convenient for what follows.

First, the eigenvalue c enters (9)–(12) *only* through the convective factor $i\alpha(\bar{U} - c)$. The right-hand operator $\mathbf{M}$ is therefore purely algebraic, containing no derivatives, and can be written down at once:

$$\mathbf{M}(\alpha; y) = i\alpha \begin{pmatrix} 0 & 0 & -1/\bar{T} & \gamma M_e^2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (15)$$

Second, only three terms in the entire system carry D^2 : the viscous diffusion of $\hat{u}$ and $\hat{v}$ in (10)–(11) and the heat conduction of $\hat{T}$ in (12). The second-order coefficient matrix is therefore sparse:

$$\mathbf{A}_2(y) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\nu_R \bar{\mu} & 0 & 0 & 0 \\ 0 & -\frac{4}{3}\nu_R \bar{\mu} & 0 & 0 \\ 0 & 0 & -\mathcal{C}_\kappa & 0 \end{pmatrix}, \quad (16)$$

where the signs follow from moving the right-hand sides of (10)–(12) to the left. The pressure amplitude $\hat{p}$ is never differentiated twice, so its column in $\mathbf{A}_2$ is empty, reflecting the absence of a pressure diffusion mechanism. The first- and zeroth-order matrices $\mathbf{A}_1$ and $\mathbf{A}_0$ are read off from (9)–(12) in the same manner; their entries are the convective, pressure-gradient, mean-shear, and transport-derivative coefficients, and they are assembled term by term in `pymack/temporal_solver.py`.

Temporal and spatial problems. Equation (14) supports the two standard closures. Fixing $\alpha \in \mathbb{R}$ leaves c as the eigenvalue of a *linear* operator pencil; this is the temporal problem. Fixing $\omega \in \mathbb{R}$ instead and regrouping the same equations by powers of α gives

$$(\mathbf{C}_0(\omega; y, D) + \alpha \mathbf{C}_1(\omega; y, D) + \alpha^2 \mathbf{C}_2(y)) \hat{q} = 0, \quad (17)$$

a *quadratic* pencil in α ; this is the spatial problem, reduced to linear form by companion linearisation in the companion *Numerical Methods* document.

The discretised derivative operator. Up to this point D has denoted the continuous derivative d/dy acting on smooth functions. Numerical solution replaces it by a discrete differentiation operator: the amplitudes are sampled on a collocation grid $\{y_j\}_{j=0}^n$, and

$$D \longrightarrow D_1, \quad D^2 \longrightarrow D_2, \quad D_1, D_2 \in \mathbb{R}^{(n+1) \times (n+1)}, \quad (18)$$

where D_1, D_2 are the Chebyshev collocation differentiation matrices mapped to the physical domain, the natural discretisation for boundary-layer eigenvalue problems [5, 4]. Under (18) each mean-flow coefficient becomes a diagonal matrix, each entry of $\mathbf{A}_2, \mathbf{A}_1, \mathbf{A}_0, \mathbf{M}$ becomes an $(n+1) \times (n+1)$ block, and the four sampled amplitudes are stacked into a single vector $\boldsymbol{\phi} = (\hat{u}; \hat{v}; \hat{T}; \hat{p}) \in \mathbb{C}^{4(n+1)}$. The operator statement (14) then becomes the finite-dimensional generalised eigenvalue problem

$$\boxed{A(\alpha) \boldsymbol{\phi} = c B(\alpha) \boldsymbol{\phi}} \quad A, B \in \mathbb{C}^{4(n+1) \times 4(n+1)}, \quad (19)$$

which is the object `pymack/temporal_solver.py` assembles and solves.

Anatomy of the discrete problem. Equation (19) deserves to be read carefully: it is the computational atom of the entire framework, and each of its symbols has a definite meaning inherited from the derivation above.

The matrix $A(\alpha)$ is the discretisation of the left-hand operator $\mathbf{L}$ of (14): it carries *all of the eigenvalue-independent physics*, namely mean-flow convection $i\alpha\bar{U}$, the pressure gradient, mean-shear production, viscous diffusion, heat conduction, and the dilatational coupling, evaluated at the chosen wavenumber α . The matrix $B(\alpha)$ is the discretisation of the algebraic operator $\mathbf{M}$ of (15): it marks exactly *where the eigenvalue enters* the equations, which, since $-i\omega = -i\alpha c$, is precisely where the time derivative acted before the ansatz (6). Both inherit the 4×4 block structure of the operator form: with i, j ranging over the rows (C, X, Y, E) and the columns ($\hat{u}, \hat{v}, \hat{T}, \hat{p}$),

$$A_{ij}(\alpha) = \Lambda([A_2]_{ij}) D_2 + \Lambda([A_1]_{ij}) D_1 + \Lambda([A_0]_{ij}), \quad B_{ij}(\alpha) = \Lambda([M]_{ij}), \quad (20)$$

where $\Lambda(f) = \text{diag}(f(y_0), \dots, f(y_n))$ places a known coefficient function on the grid. Every entry of A and B is therefore an $(n+1) \times (n+1)$ block built from exactly three ingredients: the differentiation matrices D_1, D_2 , diagonal samples of mean-flow profiles, and the scalars α, M_e, Re . The eigenvector $\phi = (\hat{u}; \hat{v}; \hat{T}; \hat{p})$ stacks the four amplitude profiles sampled on the same grid: it *is* the eigenfunction $\hat{q}(y)$ of (14), represented by its values at the collocation points.

A single solve of (19) returns the full discrete spectrum: of order $4(n+1)$ eigenpairs (c, ϕ) . Each eigenpair is a candidate normal mode of the boundary layer at the given (M_e, Re, α) : the real part c_r is its phase speed as a fraction of U_e ; the imaginary part sets its temporal growth rate $\omega_i = \alpha c_i$, so $c_i > 0$ identifies an amplifying wave, $c_i < 0$ a damped one, and $c_i = 0$ a neutral one; and the four segments of ϕ give the wave's internal structure: the velocity, temperature, and pressure amplitude profiles across the layer (a wall-peaked $|\hat{p}|$, for instance, signals a second mode). Only a handful of these eigenpairs are physical *discrete* modes; the remainder approximate the continuous spectra or are numerical artifacts. In particular, the boundary-condition rows carry no eigenvalue and leave zero rows in B , so B is singular and the pencil must be solved in generalised (QZ) form, producing spurious infinite eigenvalues that are discarded. The identification of the physical modes within this spectrum is treated in the companion *Numerical Methods* document.

Everything the framework ultimately produces is assembled from repeated solves of (19): one solve interrogates one point of the parameter space (M_e, Re, α) , a sweep of solves traces the growth rate across the (R, α) plane, the zero level set $c_i = 0$ of that sweep is the neutral curve, and the integral of the growth rate along frequency lines is the N -factor. The construction of D_1, D_2 , the wall-clustering domain map, the boundary-condition rows, the solution of (19) and of the spatial pencil (17), and the classification of the resulting spectrum are the subject of the companion *Numerical Methods* document.

5 Physical character of the instability modes

The disturbance equations admit discrete eigenmodes; a temporal solve returns a complex phase speed such as $c = 0.930 + 0.0200i$. This section summarises the physics behind the standard mode labels, so that such a number identifies a definite wave.

First (Tollmien–Schlichting) mode. At low speed the first mode is a viscous instability ($c_r \approx 0.3$ – 0.6): viscosity introduces a phase lag between u' and v' near the wall, so the Reynolds stress $-\bar{\rho} u' v'$ no longer averages to zero and extracts energy from the mean shear. As the Mach number rises, compressibility creates a generalised inflection point (below) and the first mode becomes

increasingly inviscid and inflectional, with viscosity then often acting to stabilise; it is the dominant instability only up to $M_e \approx 4$ [1].

Second (Mack) mode. Above $M_e \approx 4$ a second, usually stronger, instability appears: an acoustic wave trapped between the wall and the relative sonic line, reflecting within a near-wall acoustic waveguide [1, 11]. This interpretation accounts for its principal properties:

- its phase speed is $c_r \approx 1 - 1/M_e$, the limiting value at which the wave is sonic relative to the edge flow, which is why the spatial-solve shift target $\alpha_0 = \omega/(1 - 1/M_e)$ is effective; the realised c_r sits somewhat above this and varies along the branch (0.93 against $1 - 1/6 = 0.83$ for a representative $M_e = 6$ mode);
- it is frequency-tuned: as in any resonator the wavelength is set by the cavity thickness, so the most-amplified wavelength is $\sim 2\delta$ and the frequency scales as $f \sim U_e/2\delta$, rising as the layer thins downstream;
- its eigenfunction has a wall-peaked pressure amplitude, the acoustic field of the waveguide.

The organising quantity: relative Mach number. Both mechanisms are organised by the relative Mach number

$$M_{\text{rel}}(y) = \frac{\bar{U}(y) - c_r}{\bar{a}(y)}, \quad \bar{a} = \frac{\sqrt{T}}{M_e}, \quad (21)$$

with $\bar{a}$ the local sound speed. The critical layer is the height at which $M_{\text{rel}} = 0$, i.e. $\bar{U} = c_r$, where the eigenfunction typically peaks and the convective factor $i\alpha(\bar{U} - c)$ in (9)–(12) vanishes; the relative sonic line is $|M_{\text{rel}}| = 1$; and the second-mode acoustic waveguide is the near-wall region in which $|M_{\text{rel}}| > 1$.

The generalised inflection point. By Rayleigh’s theorem an inviscid incompressible shear flow requires a velocity inflection point ($\bar{U}'' = 0$) for instability. Lees & Lin [6] generalised this to compressible flow: the necessary condition for the inviscid first mode is a *generalised inflection point*, $(\bar{\rho}\bar{U}')' = 0$. Aerodynamic heating places a hot, low-density layer near the wall and pushes that point up into the layer, giving compressible boundary layers a robust inviscid first-mode instability that low-speed layers lack. The second mode requires no inflection point; its existence condition is instead the relative-supersonic region $|M_{\text{rel}}| > 1$ above.

The wall-cooling reversal. Cooling the wall stabilises the first mode (it weakens the near-wall viscous mechanism) but destabilises the second (it thickens the relative-supersonic region and raises the tuned frequency) [1]. Cold-walled hypersonic vehicles are therefore second-mode dominated, which is why the solver carries an isothermal-wall option.

Phase and group velocity. c_r is the phase speed; energy propagates at the group velocity $c_g = d\omega_r/d\alpha_r$, and it is c_g that rigorously links temporal to spatial growth [7] (used when the companion *Numerical Methods* document converts between the two formulations).

The following comparison places any computed eigenvalue at a glance:

	First (TS) mode	Second (Mack) mode
mechanism	viscous (low speed) $\rightarrow$ inviscid inflectional (high speed)	trapped acoustic wave (wall-sonic line)
phase speed c_r	0.3–0.6	$\approx 1 - 1/M_e$
dominant when	low speed, $M_e \lesssim 4$	high speed, $M_e \gtrsim 4$
frequency	broadband	tuned, $f \sim U_e/2\delta$
existence criterion	generalised inflection point [6]	relative-supersonic region, $ M_{\text{rel}} > 1$
wall cooling	stabilises	destabilises
most-unstable wave	oblique (3D)	2D ($\psi = 0$)

6 Assumptions and range of validity

The equations derived above are the two-dimensional, parallel-flow, modal disturbance equations. Four idealisations bound their interpretation:

1. **Parallel flow.** Setting $\bar{v} = 0$ with y -only coefficients (Step 3) neglects the $O(1/R)$ streamwise growth of the layer; the systematic remedy is the parabolised stability equations, which march the disturbance downstream retaining the slow x -variation [8].
2. **Two-dimensionality.** The ansatz takes $\beta = 0$ (Step 4). The most-unstable first mode is oblique ($\psi \sim 55^\circ$; Squire’s theorem does not hold under compressibility [1]), so two-dimensional first-mode growth is a lower bound; the second mode is most unstable at $\psi = 0$, so the two-dimensional result is its physical maximum.
3. **Modal analysis.** Eigenvalues describe asymptotic exponential growth; the non-normal operator also permits transient (algebraic) growth that modal analysis misses, relevant to bypass and blunt-body transition paths [9].
4. **Receptivity and e^N .** LST provides the amplification *ratio* $A(R)/A(R_0) = e^N$, not the initial amplitude $A(R_0)$; determining that amplitude is the separate receptivity problem, and the e^N method is an environment-dependent correlation rather than a first-principles criterion [10, 11].

None of these affect the equations themselves; they delimit their range of validity. A strong second mode is captured faithfully by the two-dimensional modal equations derived here, which is why they are the workhorse of hypersonic transition prediction [1, 11]. With the system (9)–(12) in operator form (14) and its discrete counterpart (19) identified, the construction and solution of the matrix problem proceed in the companion *Numerical Methods* document.

References

- [1] L. M. Mack, “Boundary-layer linear stability theory,” AGARD Report No. 709, *Special Course on Stability and Transition of Laminar Flow*, 1984.
- [2] F. M. White, *Viscous Fluid Flow*, 3rd ed., McGraw–Hill, New York, 2006.
- [3] S. Özgen and S. A. Kırcah, “Linear stability analysis in compressible, flat-plate boundary layers,” *Theoretical and Computational Fluid Dynamics*, Vol. 22, 2008, pp. 1–20.
- [4] M. R. Malik, “Numerical methods for hypersonic boundary layer stability,” *Journal of Computational Physics*, Vol. 86, No. 2, 1990, pp. 376–413.

- [5] S. A. Orszag, “Accurate solution of the Orr–Sommerfeld stability equation,” *Journal of Fluid Mechanics*, Vol. 50, No. 4, 1971, pp. 689–703.
- [6] L. Lees and C. C. Lin, “Investigation of the stability of the laminar boundary layer in a compressible fluid,” NACA Technical Note No. 1115, 1946.
- [7] M. Gaster, “A note on the relation between temporally-increasing and spatially-increasing disturbances in hydrodynamic stability,” *Journal of Fluid Mechanics*, Vol. 14, 1962, pp. 222–224.
- [8] T. Herbert, “Parabolized stability equations,” *Annual Review of Fluid Mechanics*, Vol. 29, 1997, pp. 245–283.
- [9] P. J. Schmid and D. S. Henningson, *Stability and Transition in Shear Flows*, Springer, New York, 2001.
- [10] J. L. van Ingen, “A suggested semi-empirical method for the calculation of the boundary layer transition region,” Report VTH-74, Delft University of Technology, 1956.
- [11] A. Fedorov, “Transition and stability of high-speed boundary layers,” *Annual Review of Fluid Mechanics*, Vol. 43, 2011, pp. 79–95.